

## LAKATRASK, Sweden

WMO: 021740

Lat: 66.2789N

Lon: 21.1285E

Elev: 185

StdP: 99.12

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -31.6      | -28.6 | -35.4                         | 0.1 | -31.5 | -32.3 | 0.2 | -28.5 | 5.6                   | -2.5 | 5.1 | -2.8 | 0.1                   | 40   | 0.656 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.0                   | 25.4            | 16.5 | 23.4 | 15.6 | 21.5 | 14.9 | 18.1                | 23.2 | 17.0 | 21.1 | 15.8 | 19.8 | 2.3                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.4                            | 11.9 | 19.6 | 15.2 | 11.0 | 18.7 | 14.1 | 10.2 | 17.4 | 51.9          | 23.0 | 48.3 | 21.2 | 44.9 | 19.6 | 22.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 5.4               | 4.6  | 4.1 | -33.6                      | 27.5 | 2.9                | 2.6 | -35.7                                              | 29.4 | -37.4      | 30.9 | -39.0      | 32.4 | -41.1      | 34.3 |    |    |
|                   |      |     | -34.0                      | 19.3 | 2.8                | 1.5 | -36.0                                              | 20.4 | -37.6      | 21.3 | -39.2      | 22.1 | -41.2      | 23.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov  | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)  | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 0.4    | -12.9 | -12.1 | -6.6  | 0.3   | 6.4   | 11.3  | 14.7  | 12.4  | 7.3   | 0.1   | -6.4 | -10.1 |
|                                                                                  | DBStd       | 10.96  | 7.85  | 7.75  | 5.73  | 3.90  | 3.84  | 3.58  | 3.01  | 3.36  | 3.55  | 4.81  | 6.45 | 7.65  |
|                                                                                  | HDD10.0     | 3819   | 712   | 620   | 513   | 292   | 125   | 24    | 2     | 14    | 95    | 307   | 493  | 622   |
|                                                                                  | HDD18.3     | 6542   | 970   | 853   | 772   | 542   | 370   | 212   | 119   | 185   | 331   | 565   | 743  | 881   |
|                                                                                  | CDD10.0     | 327    | 0     | 0     | 0     | 0     | 13    | 65    | 146   | 89    | 14    | 0     | 0    | 0     |
|                                                                                  | CDD18.3     | 9      | 0     | 0     | 0     | 0     | 0     | 2     | 6     | 2     | 0     | 0     | 0    | 0     |
|                                                                                  | CDH23.3     | 182    | 0     | 0     | 0     | 0     | 8     | 44    | 102   | 27    | 0     | 0     | 0    | 0     |
|                                                                                  | CDH26.7     | 26     | 0     | 0     | 0     | 0     | 1     | 9     | 15    | 1     | 0     | 0     | 0    | 0     |
| Wind                                                                             | WSAvg       | 1.6    | 1.1   | 1.3   | 1.7   | 1.9   | 2.1   | 2.0   | 1.8   | 1.7   | 1.7   | 1.5   | 1.2  | 1.3   |
| Precipitation                                                                    | PrecAvg     | 579    | 39    | 33    | 26    | 29    | 41    | 63    | 82    | 65    | 59    | 49    | 53   | 38    |
|                                                                                  | PrecMax     | 814    | 69    | 57    | 48    | 61    | 78    | 113   | 118   | 123   | 214   | 86    | 94   | 82    |
|                                                                                  | PrecMin     | 400    | 4     | 11    | 10    | 13    | 9     | 5     | 35    | 24    | 18    | 22    | 22   | 12    |
|                                                                                  | PrecStd     | 111    | 16    | 13    | 12    | 12    | 18    | 29    | 22    | 28    | 46    | 20    | 22   | 17    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 5.0   | 6.1   | 8.7   | 15.9  | 24.5  | 27.7  | 28.5  | 26.1  | 19.1  | 12.1  | 7.3  | 4.7   |
|                                                                                  |             | MCWB   | 3.0   | 3.3   | 4.0   | 7.9   | 13.7  | 16.8  | 18.3  | 18.9  | 13.5  | 8.4   | 5.4  | 2.6   |
|                                                                                  | 2%          | DB     | 1.5   | 3.3   | 6.1   | 11.6  | 21.3  | 23.9  | 25.9  | 23.6  | 16.8  | 10.0  | 5.3  | 2.4   |
|                                                                                  |             | MCWB   | 0.4   | 1.1   | 2.2   | 5.8   | 12.8  | 14.4  | 17.3  | 16.4  | 12.5  | 7.7   | 4.1  | 1.1   |
|                                                                                  | 5%          | DB     | -0.2  | 0.9   | 4.3   | 9.6   | 17.6  | 21.5  | 24.0  | 21.6  | 15.2  | 8.4   | 3.1  | 1.0   |
|                                                                                  |             | MCWB   | -0.7  | -0.3  | 1.3   | 4.5   | 10.6  | 13.5  | 16.5  | 15.7  | 11.6  | 6.9   | 2.5  | 0.3   |
|                                                                                  | 10%         | DB     | -2.1  | -0.8  | 2.4   | 7.6   | 14.7  | 19.2  | 22.0  | 19.4  | 13.6  | 6.8   | 1.6  | 0.1   |
|                                                                                  |             | MCWB   | -2.6  | -1.6  | -0.1  | 3.5   | 8.6   | 12.5  | 15.7  | 14.7  | 10.9  | 5.5   | 1.1  | -0.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.2   | 3.6   | 4.2   | 8.2   | 15.0  | 17.7  | 19.8  | 19.8  | 14.5  | 9.5   | 5.7  | 2.9   |
|                                                                                  |             | MCDB   | 5.0   | 6.0   | 8.2   | 14.4  | 22.4  | 24.5  | 25.2  | 23.9  | 17.1  | 10.8  | 6.5  | 4.7   |
|                                                                                  | 2%          | WB     | 0.6   | 1.1   | 2.7   | 6.3   | 13.2  | 16.1  | 18.6  | 17.8  | 13.3  | 8.2   | 4.0  | 1.4   |
|                                                                                  |             | MCDB   | 1.2   | 2.7   | 5.4   | 10.9  | 20.3  | 21.8  | 23.9  | 21.9  | 15.8  | 9.6   | 5.0  | 2.2   |
|                                                                                  | 5%          | WB     | -0.7  | -0.2  | 1.5   | 4.9   | 11.1  | 14.5  | 17.5  | 16.5  | 12.4  | 6.9   | 2.4  | 0.4   |
|                                                                                  |             | MCDB   | -0.1  | 0.8   | 4.0   | 9.1   | 17.5  | 19.5  | 22.1  | 19.9  | 14.3  | 8.2   | 3.1  | 1.0   |
|                                                                                  | 10%         | WB     | -2.5  | -1.6  | 0.2   | 3.6   | 9.2   | 13.1  | 16.4  | 15.2  | 11.5  | 5.6   | 1.1  | -0.4  |
|                                                                                  |             | MCDB   | -2.1  | -0.7  | 2.0   | 7.3   | 13.8  | 17.8  | 20.4  | 18.4  | 13.1  | 6.7   | 1.6  | 0.2   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.4   | 10.3  | 11.8  | 10.3  | 11.3  | 11.0  | 11.0  | 10.8  | 9.6   | 7.5   | 7.0  | 8.1   |
|                                                                                  |             | MCDBR  | 7.2   | 9.3   | 11.4  | 13.4  | 17.3  | 15.6  | 14.9  | 14.0  | 12.4  | 9.0   | 6.5  | 6.7   |
|                                                                                  |             | MCWBR  | 6.7   | 8.0   | 8.3   | 8.0   | 9.5   | 7.4   | 7.2   | 7.7   | 8.3   | 6.8   | 5.8  | 6.3   |
|                                                                                  | 5% WB       | MCDBR  | 7.2   | 8.6   | 10.9  | 12.6  | 16.3  | 13.1  | 12.4  | 11.7  | 9.7   | 7.8   | 6.1  | 6.8   |
|                                                                                  |             | MCWBR  | 6.8   | 7.6   | 8.2   | 7.7   | 9.2   | 6.8   | 6.9   | 7.0   | 6.9   | 6.2   | 5.6  | 6.5   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.188  | 0.257 | 0.290 | 0.322 | 0.336 | 0.339 | 0.350 | 0.334 | 0.304 | 0.282 | 0.338 | N/A  |       |
|                                                                                  | taud        | 1.948  | 2.154 | 2.230 | 2.308 | 2.469 | 2.487 | 2.470 | 2.501 | 2.564 | 2.403 | 2.720 | N/A  |       |
|                                                                                  | Ebn at Noon | 242    | 602   | 771   | 841   | 869   | 872   | 848   | 827   | 778   | 591   | 257   | N/A  |       |
|                                                                                  | Edh at Noon | 16     | 52    | 80    | 96    | 92    | 93    | 92    | 80    | 60    | 41    | 17    | N/A  |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.08   | 0.65  | 2.18  | 4.05  | 4.96  | 5.23  | 4.86  | 3.60  | 2.03  | 0.74  | 0.15  | 0.01 |       |
|                                                                                  | RadStd      | 0.01   | 0.09  | 0.17  | 0.20  | 0.50  | 0.41  | 0.47  | 0.33  | 0.29  | 0.12  | 0.02  | 0.00 |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (38 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page